

LAURA CALIGARIS

Bioinformatician with a background in microbial ecology and metagenomics, and prior experience in structural bioinformatics from my MSc. Proficient in R for statistical and ecological data analysis, with working knowledge of Python and hands-on experience with bioinformatic pipelines on HPC infrastructure. Experience integrating multi-domain sequencing data (prokaryotic and eukaryotic) and working in international research environments.

PERSONAL DATA

EMAIL:	caligaris.laura15@gmail.com	IN:	in/LauraCaligaris
PHONE:	(+39) 340-3152534	GITHUB:	github/LauraCaligaris

SCIENTIFIC EDUCATION

04/2027	Doctorate degree , Earth, Life and Environment Sciences University of Bologna Thesis: <i>NGS and Metagenomic applications for in situ evaluation of water supplies contamination according to the water safety plan</i>
11/2023–11/2026	PhD student at University of Bologna
11/2023	Master's Degree in Bioinformatics University of Rome Tor Vergata Thesis: <i>Identification of potential inhibitors of human topoisomerase IB through virtual screening and classical molecular dynamics techniques</i>
10/2020–11/2023	Studies of Bioinformatics at University of Rome Tor Vergata
2/2020	Bachelor's Degree in Biological Sciences University of Pisa Thesis: <i>Function and Structure of PET Hydrolase (PETase): an enzyme for plastic's biodegradation</i>
2013–2020	Studies of Biological Sciences at University of Pisa

EXPERIENCE

11/2023–Present <i>Intern</i>	PhD Student in Earth, Life and Environment Sciences University of Bologna
10/2025–04/2026 <i>Visiting Researcher</i>	Study of taxonomic, ecological and functional structure of water biofilms in taps through metagenomics data and interaction networks Swiss Federal Institute of Aquatic Science and Technology, Dübendorf (Switzerland)
06/2022–10/2023 <i>Intern</i>	MPXV and Human Topoisomerase Virtual Screening University of Rome Tor Vergata
11/2022–01/2023 <i>Intern</i>	Analysis of drugs on MPXV and Human Topoisomerase: activity assay, religation and relaxation assays University of Rome Tor Vergata
05/2017–07/2017 <i>Intern</i>	Antibody analysis in autoimmune diseases University of Pisa

MENTORSHIP AND SERVICE

16–18/02/2026 <i>Tutor</i>	Microbiology Praktikum ETH Zürich Support to students on basic microbiology techniques to cultivate and select bacterial species.
11/2022–06/2023 <i>Tutor</i>	Structural Bioinformatics Course University of Rome Tor Vergata Support to students during practical lectures involving the use of the main bioinformatics software and servers.

TECHNICAL SKILLS

PROGRAMMING	Proficient in R, working knowledge in Python, Ruby and C. Basic knowledge of database management systems, MySQL and the SQL language. Familiar with HTML5, CSS, PHP.
(META)GENOMICS	Assembly, quality and annotation tools (FastQC, Trimmomatic, SPAdes, Pilon, Prokka, BUSCO, ABRicate, Snippy, PGGB, CheckM2); alignment tools (MUMmer, BWA, Bowtie2, SAMtools, BCFtools); metagenomics and amplicon sequencing analysis (Kraken2, mothur, QIIME 2, DADA2, phyloseq); taxonomy (FastTree, PICRUSt2, iTOL, PR2, GTDB); co-occurrence network inference (SPIEC-EASI).
STATISTICS	Multivariate community ecology statistics (PERMANOVA/adonis2, Procrustes, Mantel test, Spearman correlation); alpha and beta diversity metrics (Shannon, observed richness, Chao1, Bray-Curtis, Jaccard); data visualization and figure production (ggplot2, patchwork, vegan).
STRUCTURAL	Virtual screening and drug design (AutoDock Vina); molecular dynamics and trajectory analysis (AMBER, GROMACS); processing of simulation data with Python and Bash scripting.
VISUALIZATION	PyMOL, Chimera, VMD, Krona, iTOL.
DATABASES	NCBI and scientific databases (PubMed, GenBank, PubChem, PDB, UniProt, PR2, GTDB).
LINUX & HPC	Bash scripting for data processing, basic Linux administration tools and text manipulation commands. Work on computing clusters with resource optimization through the SLURM scheduler. Conda and Conda environments.

WET LAB SKILLS

LEGIONELLA	Isolation, culture and typing of <i>Legionella</i> spp.
MALDI-TOF MS	Analysis with the MALDI Biotyper® System (Bruker); construction of dendrograms; implementation of in-house databases.
MICROBIOLOGY	Preparation of culture media for the growth of prokaryotic cells.
MOLECULAR BIOLOGY	Extraction of prokaryotic DNA with standard methods or specific kits; quantification of nucleic acids (spectrophotometric methods, Qubit/NanoDrop).

SOFT SKILLS

PROBLEM SOLVING	Able to quickly adapt to project changes and reorganize work.
COMMUNICATION	Scientific presentations, exhibitions and workshop organization.
TEAMWORK	Able to manage conflicts and work as part of a team.
MANAGEMENT	Able to manage time and work and meet deadlines.

SELECTED COURSEWORK

08-09/10/2024 **Computational Pangenomics**, offered by Dr. Andrea Guarracino, Alta Formazione Insubria;
05-07/02/2024 **Introduction to Python Programming**, offered by Cineca HPC Department;
04-06/12/2023 **High Performance Bioinformatics**, offered by Cineca HPC Department;

SELECTED PUBLICATIONS AND POSTERS

Caligaris L., Girolamini L., Bonincontro C., Derelitto C., Lanzoni L., Sabelli C., Cristino S. Metabarcoding description of biofilm bacterial communities in drinking water outlets across a disinfection gradient in healthcare and industrial settings. Under review in **Microbial Ecology, Collection Biofilms and Surface-associated Microbial Communities**.

Cristino S., **Caligaris L.**, Salaris S., Derelitto C., Bonincontro C., Marino F., Grottola A., Girolamini L. *Legionella petroniana* sp. nov., a novel species isolated in Bologna, Italy: taxonomic, genomic and ecological insights in the era of environmental change. **Int. j. Environ. Res. Public Health** 2026, 23(2), 174; [doi: 10.3390/ijerph23020174](https://doi.org/10.3390/ijerph23020174).

Caligaris L., Hammes F., Cristino S., Gabrielli M. Coupled 16S and 18S rRNA analysis reveals complex *Legionella*-amoeba associations in non-disinfected shower hose biofilms. **11th International Conference on Legionella**, Zurich, Switzerland, 7-9 September 2026.

Caligaris L., Girolamini L., Bonincontro C., Derelitto C., Sabelli C., Cristino S. Microbiome profiling using 16S metagenomics approach to improve water quality assessment. **ASM Microbe 2025, American Society for Microbiology**, Los Angeles (CA), United States, 19-23 June 2025.

LANGUAGES

ITALIAN	Native
ENGLISH	Advanced
FRENCH	Basic